LAWS OF PRODUCTION

The laws of production describe the technically possible ways of increasing the level of production. output may increase in various ways

Output can be increased by changing all factors of production. Clearly this is possible only in the long run. Thus, the laws of return to scale refer to the long-run analysis of production.

In the short run output may be increased by using more of the variable factor(s), while capital (and possibly other factors as well) are kept constant. The marginal product of the variable factors will decline eventually as more and more quantities of this factor are combine with other constant factors. The expansion of output with one factor (at least) constant is described by the law of (eventually) diminishing returns of the variable factor, which is often referred to as the law of variable proportions

LAWS OF RETURN TO SCALE

The laws of return to scale refer to the effects of scale relationships.

In the long run output may be increased by changing all factors by the same proportion or by different proportions.

Traditional theory of production concentrates on the first case, that is, the study of output as all input changes by the same proportion. The term 'return to scale' of the changes in output as all factors change by the same proportions.

Production Isoquant

An Isoquant includes (is the locus of) all the technically efficient methods (or all the combinations of factors of production) for producing a given level of output.

The production Isoquant may assume various ~~shape~~ shapes depending on the degree of substitutability of factors.

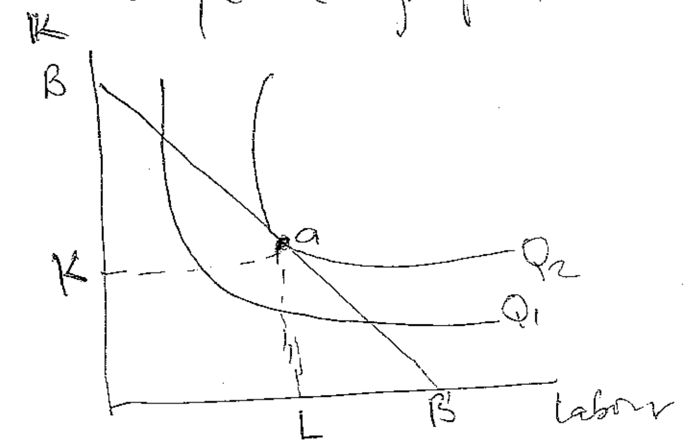
- Linear Isoquant. This type assumes perfect substitutability of factors of production: a given commodity may be produced by using only capital or only labour.

- Input-output Isoquant. This assumes strict complementarity (that is, zero substitutability) of the factors of production. There is only one method of production for any one commodity.

- Kinked Isoquant. This assumes limited substitutability of K and L. There are only a few process for producing any one commodity. Substitutability of the factors is possible only at the kinks. This is basically used in linear programming.

Smooth, convex Isoquant. This form assumes (B) continuous substitutability of K and L only over a certain range, beyond which factors cannot substitute each other. The Isoquant appears as a smooth curve convex to the origin.

Isoquant graph



Optimal production level

from the above graph line B, B is isocost line, which shows the entire alternative combinations of factors (capital and labour) a firm can purchase for a given expenditure

The optimal input combination occurs at point a, the point of tangency between the isocost line and a production isoquant curves Q_2

MARKET STRUCTURE

Market structure describes how a market is organised in terms of how much competition exists, usually on the supply side. Also, it refers to the physical characteristics of the market within which firms interact.

Market structures are classified into two broad categories which are;

- Perfect market
- Imperfect market

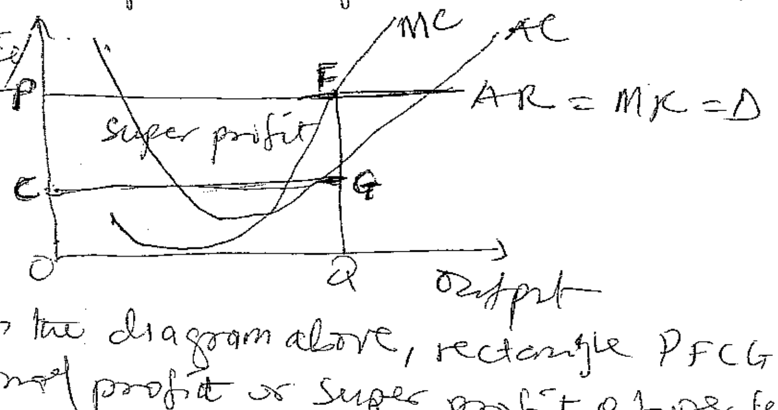
Perfect market

A perfect market is one in which there is perfect competition. Competition ordinarily means rivalry among firms. Perfect competition therefore refers to the market situation in which there exist a large number of buyers and sellers who cannot individually influence the price of commodities. Markets for coal and farm product are good examples of perfectly competative markets.

Conditions necessary for perfect competition

(a) large numbers of buyers and sellers
(b) Homogeneous commodities
(c) There is free entry and exit of buyers and sellers
(d) Adequate information
(e) portability of goods
(f) There is no preferential treatment
(g) Both buyers and sellers are price takers

Short-run profit maximising profit for a perfectly competitive firm

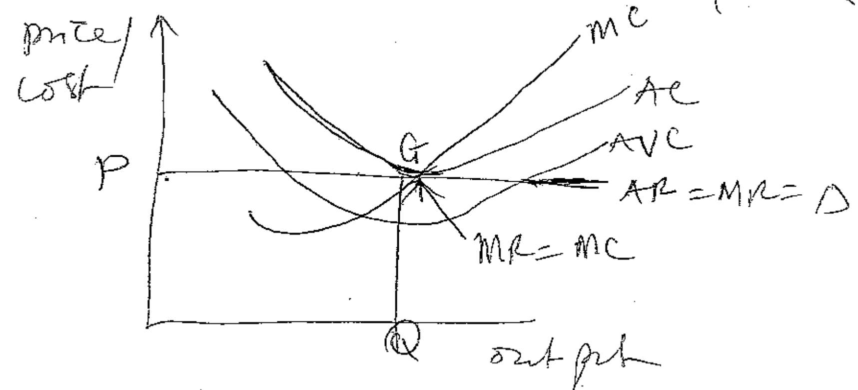


In the diagram above, rectangle PFCG represents the abnormal profit or super profit of perfect competitor in shortrun. The firms maximised profit by producing at a cost (C) below the market ruling price (P) and where MR = MC. The market price is P and the firms output cost of production at this price are Q and C respectively.

long-run equilibrium of a perfectively competitive firm

Recall that one of the conditions of the existence of a perfect market is that there must be free entry and exit by firms into and out of the market. With this condition, the super profit (or abnormal) profit made in the short-run will attract many firms into the market in order to grab a share of the abnormal profit. This process of new entry will continue until the super profit of each firm in the industry or market drops to zero or is completely eliminated. This stems from excess supply and continous fall in price to the level where price equals the average total cost, which eventually leads to the earning of only normal profit in the long run

This is represented graphically below:



Normal profit is often earned in the long-run and it is maximised where MC = MR. Normal profit is regarded as the level of profit required to keep a firms in production.

Monopoly

Monopoly is a market structure where there is only one firm producing a commodity which has no close substitutes and with barriers limiting other firms from entering the industry or market.

The main causes that lead to monopoly are the following

(i) ownership of strategic raw materials or exclusive knowledge of production techniques

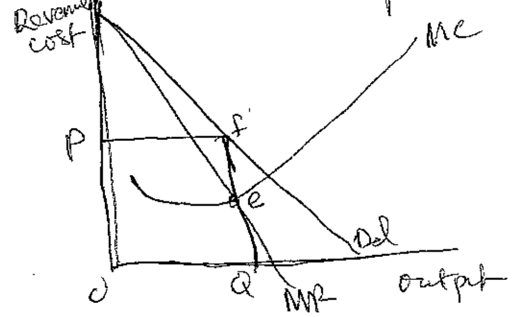
(ii) patent rights for a product

(iii) government licensing or the imposition of foreign trade barriers to exclude foreign competitors

(iv) the size of the market.

Short run equilibrium and profit position ~~of~~ a monopoly

Pure monopoly is the polar opposite of perfect competition on the market structure. An absence of competitors permits the monopoly firm some control over the market price by allowing it vary the amounts of product offered for sale. This follows the fact that monopoly firms faces a downward-sloping demand curve, meaning, the monopolist can restricts output to increase profits and consumer will pay a higher price if fewer units are placed on the market.



When the firm faces a downward-sloping demand curve, the firm's marginal revenue (MR) curve lies below it. To maximize profits, the monopoly firm determine output level by setting MC = ~~AR~~ MR at point 'e' at this point the profit maximizing conditions are fulfilled. it charges what the market will bear, as indicated by point 'f' on the demand curve directly above point 'e'. (9)

Oligopoly Market Structure

This is a market situation where there are a few sellers of homogenous or differentiated products. The firms are mutually independents. That is to say each firm in making its production and sale policy or decisions considered the effect of such decisions on other firms in the industry because any change in output or price by one firm will substantially affect the sale of competing ~~price by one~~ or rival firms. For example no single petroleum company in Nigerian market can make a change in its pricing and sales without direct repercussion on the sales of its competitors.

Pricing policies and practices

Pricing is the exchange money value of product or services. it is the amount of money buyers are willing to pay and sellers are willing to accept for goods or services under consideration for exchange. As a general rule, a point at which ~~specific~~. supply and demand curve intersect, defines the price for specific good.

For example
suppose the demand function for product is given as $Qd = 42 - 2P$
and supply function is given as:

$Qs = 12 + 4P$

find the price which quantity demand is equal quantity supply (8)

$Qd = 42 - 2P$ (1)

$Qs = 12 + 4P$ (2)

$Qd = Qs$ (3)

Substitute the functions

$42 - 2P = 12 + 4P$ (4)

$-2P - 4P = 12 - 42$ (5)

$-6P = -30$ (6)

$P = \frac{-30}{-6} = 5$

The price at which the quantity demand equal to what is supply 5

What are the quantity demand and quantity supply

Substitute $P = 5$ in equation (1) & (2)

$Qd = 42 - 2(5)$

$Qd = 42 - 10$

$Qd = 32$

$Qs = 12 + 4P$

$Qs = 12 + 4(5)$

$Qs = 12 + 20$

$Qs = 32$

$Qd = 32$

$Qs = 32$

$P = 5$

Graphically